

# João Vitor Bermal Santaniello

Software Engineer · Geospatial Data, Full-Stack and E-commerce

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## SUMMARY

Software engineer with six years of production experience in water utilities, energy, healthcare and telecom. I build Python and GeoPandas data pipelines, ArcGIS Enterprise dashboards, FastAPI and Node.js services and React/Next.js applications. For Aegea, one of the largest sanitation groups in Brazil, I cut a recurring geospatial workflow from 3+ hours to about 20 minutes and built the analyses behind its expansion studies. MBA in Software Engineering (USP/ESALQ, thesis 9/10). Fluent English with US teams.

## EXPERIENCE

**Software Developer**, Bizpoke Soluções em Software · São José dos Campos, SP    Dec 2022 to Present

- Cut a recurring geospatial workflow at Aegea from more than 3 hours to about 20 minutes by building a reusable Python and GeoPandas extraction tool with parallel spatial joins, validation and Excel/Shapefile export.
- Delivered census tract indicators for 16 Brazilian states by crossing IBGE, CNEFE and ANEEL BDGD data, supporting water and sewer concession studies.
- Co-designed the CITIUS/Regenera landfill biogas data model and loaded its historical data, feeding ArcGIS dashboards, web maps and heatmaps used by operations.
- Kept production dashboards running through the ArcGIS Enterprise 11.5 migration and automated weekly routines with ArcGIS toolboxes, Python, FME and PostgreSQL.
- Led front-end and mobile delivery for DOMO (Equatorial Energia): offline map editing in React Native plus Next.js, TypeScript and Redux screens designed in Figma.
- Built OCR exam processing (AWS Textract, S3) for teleophthalmology and ISPDrive, a SaaS for internet providers (Next.js, FastAPI, DigitalOcean Spaces).

**AI Coding Benchmark Contributor (freelance)**, Alignerr · Remote    May 2025 to Present

- Wrote Python problems, reference solutions and test suites for LiveCodeBench and BigCodeBench, each designed to break at least 2 of 4 frontier code models.

**Software Development Intern**, Bizpoke Soluções em Software · São José dos Campos, SP    Apr 2021 to Dec 2022

- Shipped React, Node.js, Python and C# features and produced spatial data with ArcGIS Pro and Enterprise alongside offshore teams.

## EDUCATION

**MBA in Software Engineering**, Universidade de São Paulo (USP/ESALQ)    2024 to 2026  
Thesis graded 9/10: deep learning pipeline for retinal images (Dice 0.827, A/V macro F1 0.963).

**Associate degree in Systems Analysis and Development**, Universidade Paulista (UNIP)    2021 to 2022

**Technical Program in Computing, with high school**, Colégio Técnico Antônio Teixeira Fernandes    2018 to 2020

## SKILLS

- Languages:** Python, TypeScript, JavaScript, SQL, Bash
- Data and GIS:** GeoPandas, ArcGIS Pro, ArcGIS Enterprise 11.5, ArcGIS Dashboards, Experience Builder, ArcGIS JS API, FME, PostgreSQL, enterprise geodatabases, ETL and data modeling
- Back end:** FastAPI, Node.js, REST APIs, JWT, OAuth, WebSockets, MongoDB, PostgreSQL
- Front end and mobile:** React, Next.js, Redux Toolkit, Tailwind CSS, MUI, React Native (Expo), Figma
- Cloud and DevOps:** Docker, Docker Compose, NGINX, Linux, AWS (S3, Textract, EC2), DigitalOcean Spaces, Vercel, Kubernetes (basic)
- AI and ML:** PyTorch, OpenCV, NumPy, Albumentations, OCR, LLM evaluation, AI coding agents (Claude Code, MCP)

## LANGUAGES

Portuguese (native) · English (fluent, daily professional use)